

Part 2 Risk

CHAPTER

8

The Capital Asset Pricing Model

In Chapter 7, we began to come to grips with the problem of measuring risk. Here is the story so far.

The stock market is risky because there is a wide spread of possible future rates of return. This spread is measured by the standard deviation or variance of return. The risk of any stock can be broken down into the *specific* or *diversifiable risk* that is peculiar to that stock, and the *systematic* or *market risk* from market-wide fluctuations. Investors can eliminate specific risk by holding a well-diversified portfolio, but they cannot eliminate market risk. All the risk of a fully diversified portfolio is market risk.

We showed how investors can build efficient portfolios, which offer the highest expected return for each level of portfolio risk (and also the lowest risk for each level of expected return). Investors can combine investment in risky stock portfolios with borrowing and lending. Then there is one efficient stock portfolio that is better than all the others. It is the “tangent” portfolio with the highest Sharpe ratio—that is, the efficient portfolio with the highest expected risk premium per unit of standard deviation. If all investors share the same information, they will all hold this tangent portfolio. In this case, all investors will hold the market portfolio of all stocks.

In this chapter, we build on this newfound knowledge.

Section 8.1 Market risk is measured by beta

A stock's market risk is measured by its beta. The average beta of all stocks is 1.0. Returns on stocks with betas greater than 1.0 tend to amplify market returns. Returns on stocks with betas less than 1.0 tend to move less than the market. The beta of a portfolio equals the weighted-average beta of the stocks in the portfolio.

Section 8.2 The relation between risk and return

The capital asset pricing model (CAPM) states that the expected risk premium offered by a stock equals its beta multiplied by the expected market risk premium. The CAPM was a breakthrough theory that is widely used in corporate finance. (Chapter 9 explores how the CAPM can be used to estimate the cost of capital.)

Section 8.3 Does the CAPM hold in the real world?

The CAPM “works” in several important respects, but average returns on high-beta stocks have been less than the CAPM predicts. Average returns on small-cap and value stocks have been more than the CAPM

predicts.

Section 8.4 Some alternative theories

The main competing theory is arbitrage pricing theory (APT), which relates expected risk premiums to two or more macroeconomic factors. The Fama-French three-factor model is a widely used example of APT.

8-1 Market Risk Is Measured by Beta

Chapter 7 built up to the following punchline: *The risk that a stock contributes to a well-diversified portfolio is its market risk.* Tattoo that statement on your forehead if you can't remember it any other way. It is one of the most important ideas in this book.

So how do we measure market risk? Let's go back to the definition: Market risk is the risk that a stock shares with the market. We measure it by estimating how sensitive a stock is to market movements—how much its price changes when the market moves up or down. This sensitivity is called **beta** (β), which is defined as:

Page 223

$$\beta_i = \sigma_{iM} / \sigma_M^2 \quad (8.1)$$

Here σ_{iM} is the *covariance* between the stock's return and the market's return and σ_M^2 is the variance of the market's return.

What's the intuition behind this definition? By now you should understand that the market risk of a stock depends on how it covaries with the market. So the covariance σ_{iM} in the numerator makes sense. But why do we scale it by σ_M^2 ? There are two ways to see this.

The first is a simple decomposition. To find the market risk of security i , we decompose its total risk as measured by its standard deviation into its market risk and specific risk:

$$\begin{aligned} \sigma_i &= \rho_{iM}\sigma_i + (1 - \rho_{iM})\sigma_i \\ \text{Total risk} &= \text{market risk} + \text{specific risk} \end{aligned}$$

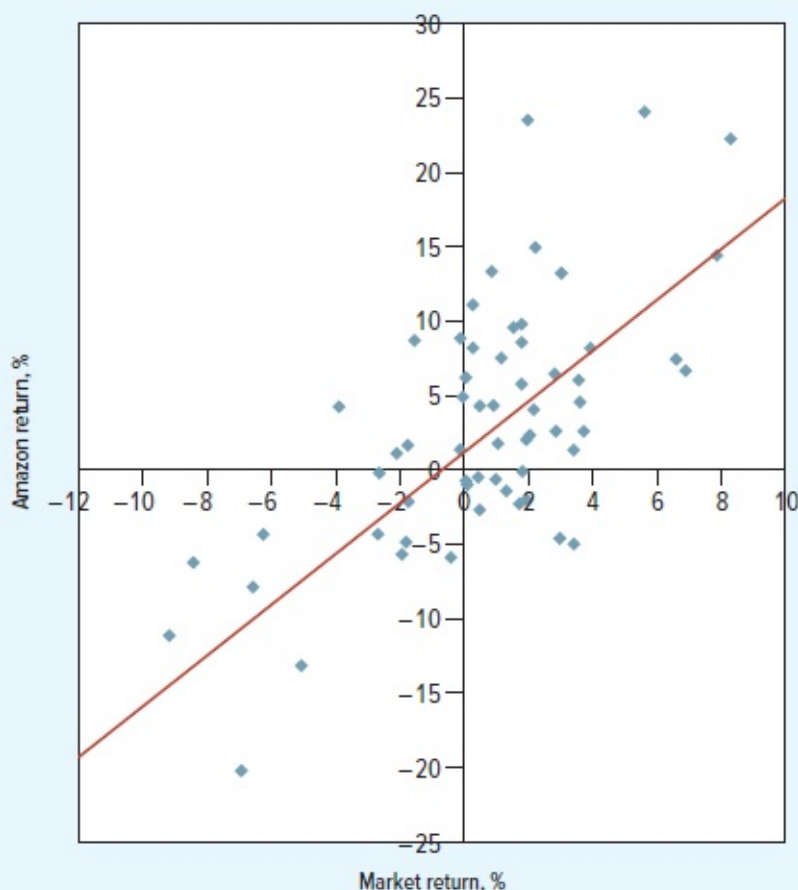
Market risk is $\rho_{iM}\sigma_i$ because the correlation coefficient ρ_{iM} measures the part of total risk σ_i that comes from the market portfolio. Recall from Eq. (7.3) that the correlation coefficient between two returns is the covariance divided by the product of the individual standard deviations. This gives us:

$$\text{Market risk} = \rho_{iM}\sigma_i = \frac{\sigma_{iM}}{\sigma_i\sigma_M}\sigma_i = \frac{\sigma_{iM}}{\sigma_M}\sigma_M = \beta_i\sigma_M \quad (8.2)$$

where $\beta_i = \sigma_{iM} / \sigma_M^2$.

The second way to explain Eq. (8.1) is by what it means. Take any month, and plot the return of Amazon that month on the y-axis and the return of the market on the x-axis. Repeat

this many times for many different months. You'll get a scatter plot as shown in Figure 8.1.



► **FIGURE 8.1**

The return on Amazon stock changes on average by 1.55% for each additional 1% change in the market return. Beta is therefore 1.55.

You can draw a best-fit line between the individual dots to show the overall relationship between Amazon's return and the market return. You don't need to draw the best-fit line by eye, but can calculate its slope by running a regression of Amazon's return on the market's return:

$$r_i = \alpha + \beta_i r_m + \varepsilon_i$$

As you may know from statistics, the best estimate of β_i —the regression *coefficient* that captures the average effect of the market return on Amazon's return—is the covariance divided by the variance, $\beta_i = \sigma_{im} / \sigma_m^2$. So, just as we wrote at the start of this section, beta measures the sensitivity of a stock's return to the market's return - how much a company's stock return goes up, *on average*, when the market goes up by 1%.

Figure 8.1 shows that, over the five years from March 2015 to February 2020, the best fit line for Amazon has a slope of 1.55—Amazon's beta was 1.55. If the future resembles the

past, then when the market rises an extra 1%, Amazon’s stock price will rise by an extra 1.55% *on average*. When the market falls an extra 2%, Amazon’s stock price will fall on average by an extra $2 \times 1.55 = 3.1\%$.

Of course, Amazon’s stock returns are not perfectly correlated with market returns. The company is also subject to specific risk, so the actual returns are scattered about the line in Figure 8.1. Sometimes, Amazon will head south while the market goes north, and vice versa. While beta measures the *slope* of the best fit line, an alternative variable, R-squared, measures the *goodness of fit* of the best fit line—how close the returns are to the line, or how little specific risk there is. R-squared is calculated as $R^2 = \sigma_{im}^2 / \sigma_i^2 \sigma_m^2$, or alternatively the square of the correlation coefficient ρ_{im} .

Let’s get back to betas, and now discuss what they tell us about how stocks behave. Stocks with betas more than 1.0 move more than one-for-one with the overall market. Examples include “luxury” companies such as Porsche. When the economy does well, consumers can buy luxury goods, but they scale back when times are tight. Stocks with betas between 0 and 1.0 still go up with the market, but less than one-for-one. Examples include “necessity” companies, such as Con Edison. In a boom, companies will produce more goods and use more electricity, so the beta is still positive. However, this sensitivity will be low. When bankers get bonuses in economic upswings, they’re more likely to celebrate it by buying a Porsche than by turning on the lights for longer.

In theory, betas can be negative. Such a stock does well when the economy does badly, such an insolvency company that helps restructure businesses that go bankrupt. Perhaps you might also say a baseball bat company, because in downturns, borrowers can’t pay back lenders, and so loan sharks might hire bailiffs to go round with baseball bats to seize the borrowers’ property. In practice, finding such companies is difficult—insolvency practices are typically part of accountancy firms who have consulting businesses that do well in upswings, and are not traded anyway.

Of course, the market is the portfolio of all stocks, so the “average” stock has a beta of 1.0. Table 8.1 reports betas for the 10 well-known stocks discussed in Chapter 7.¹ By considering what business each company is in, we can sanity check whether it makes sense if the beta is above or below 1.

Stock	Beta (β)	Stock	Beta (β)
United States Steel	2.98	Johnson & Johnson	0.75
Southwest Airlines	1.58	Tesla	0.50
Amazon	1.55	Coca-Cola	0.46
Wells Fargo	1.14	Consolidated Edison	0.31
ExxonMobil	1.14	Newmont	0.16

TABLE 8.1 Estimated betas for selected U.S. stocks, March 2015– February 2020.

Of the 10 stocks in Table 8.1, U.S. Steel has the highest beta of 2.98. It also won the prize for the highest standard deviation of 76.4% in Table 7.3 from the previous chapter. Newmont Mining is at the other extreme. Its beta is only 0.16, which may well be a surprise, given its third-place standard deviation of 36.7% in Table 7.3. The combination of high standard deviation and low beta means that most of Newmont's risk was specific and diversifiable, not systematic market risk. Newmont was a dangerous stock held alone, but a relatively safe addition to a diversified portfolio.

BEYOND THE PAGE



COVID and the risk of travel company stocks

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Just as we can measure how the returns of U.S. stocks are affected by fluctuations in the U.S. market, so we can measure how stocks in other countries are affected by movements in *their* markets. Table 8.2 shows betas for the sample of stocks from other countries.

Stock	Beta (β)	Stock	Beta (β)
Tata Motors (India)	1.83	Sony (Japan)	0.82
Hyundai Heavy Industry (Korea)	1.44	BHP Billiton (Australia)	0.80
BP (UK)	1.33	Agricultural Bank (China)	0.74
Siemens (Germany)	1.21	Sanofi (France)	0.55
Toronto Dominion Bank (Canada)	0.89	Nestlé (Switzerland)	0.11

TABLE 8.2 Betas for selected foreign stocks, March 2015–February 2020 (beta is measured relative to the stock's home market).

8.1 Self-Test

Sealink has a covariance with the market of 0.03. Its standard deviation is 0.30 (30%), and the standard deviation of the market is 0.20 (20%). What is Sealink's beta?

EXAMPLE 8.1 • Calculating Beta—A Simple Example

When financial analysts estimate beta, they generally obtain past weekly or monthly rates of

return for the stock and a market index and then deploy a statistical package to determine the slope of a line relating the stock and market returns. For Amazon, the slope was $\beta = 1.55$.

It may be helpful to see what’s going on inside those statistical packages. The following numerical example illustrates beta calculations step by step. Columns 2 and 3 in Table 8.3 show the returns over a six-month period on the market index and the stock of the Anchovy Queen restaurant chain. Although both investments provided an average return of 2%, Anchovy Queen’s stock was particularly sensitive to market movements, rising more when the market rose and falling more when it fell.

Columns 4 and 5 show the deviations of each month’s return from the average. To calculate the market variance, we need to average the squared deviations of the market returns (column 6). And to calculate the covariance between the stock returns and the market, we need to average the product of the two deviations (column 7). Beta is the ratio of the covariance to the market variance, or $76/50.67 = 1.50$. A diversified portfolio of stocks with the same beta as Anchovy Queen would be 1.5 times as volatile as the market.

BEYOND THE PAGE



Try It! Table 8.3: Calculating Anchovy Queen’s beta

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1	(1)	(2)	(3)	(4)	(5)	(6)	(7)
2							Product of
3				Deviation	Deviation	Squared	Deviations
4				from	from Average	Deviation	from Average
5		Market	Anchovy Q	Average	Anchovy Q	from Average	Returns
6	Month	Market	Anchovy Q	Deviation	Deviation	Squared	Product of
		Return	Return	from Market	from Average	Deviation from	Deviations from
				Return	Anchovy Q	Average Market	Average Returns
					Return	Return	(cols 4 × 5)
7	1	−8%	−11%	−10	−13	100	130
8	2	4	8	2	6	4	12
9	3	12	19	10	17	100	170
10	4	−6	−13	−8	−15	64	120
11	5	2	3	0	1	0	0
12	6	8	6	6	4	36	24
13	Average	2	2		Total	304	456
14				Variance = $\sigma_M^2 = 304/6 = 50.67$			
15				Covariance = $\sigma_{iM} = 456/6 = 76$			
16				Beta (β) = $\sigma_{iM}/\sigma_M^2 = 76/50.67 = 1.5$			

TABLE 8.3 Calculating the variance of the market returns and the covariance between the returns on the market and returns on Anchovy Queen. Beta is the ratio of the covariance to the variance ($\beta = \sigma_{iM}/\sigma_M^2$).

The Market Portfolio

We should pause to clarify what is meant by “market portfolio.” Section 7-5 explained that all investors who share the same information will hold the same portfolio of stocks; since all stocks must be held, all investors must hold the market portfolio.

But that discussion concerned only stocks. In reality, investors can invest in many types of risky assets, including bonds, commodities, and real estate, not just in the investor’s home country, but around the world. They can also invest in nontradable assets such as private equity. Institutional investors such as pension funds, sovereign wealth funds, and university endowments hold all these asset classes. The true market portfolio contains all the world’s risky assets, not just U.S. stocks.

In practice, there’s no index that measures the value of all risky assets. So, in practice, investors often use a *proxy*—an approximation—for the market portfolio. Then, the beta of an asset (stock, bond, etc.) is an estimate of how much that asset moves with the proxy for the market portfolio. Most investors’ portfolios are concentrated in their home countries’ stocks, so most investors use a domestic index as the market proxy. For example, U.S. investors use a U.S. stock index, such as the S&P Composite Index, as their benchmark. Investors in the

other large stock markets, including Japan, China, and the U.K., also generally measure betas relative to their own markets. Investors in smaller countries are more likely to use regional indexes. For example, an investor living in Liechtenstein would probably define beta relative to the European stock market as a whole. We return to these international topics in Chapter 28.

Why Betas Determine Portfolio Risk

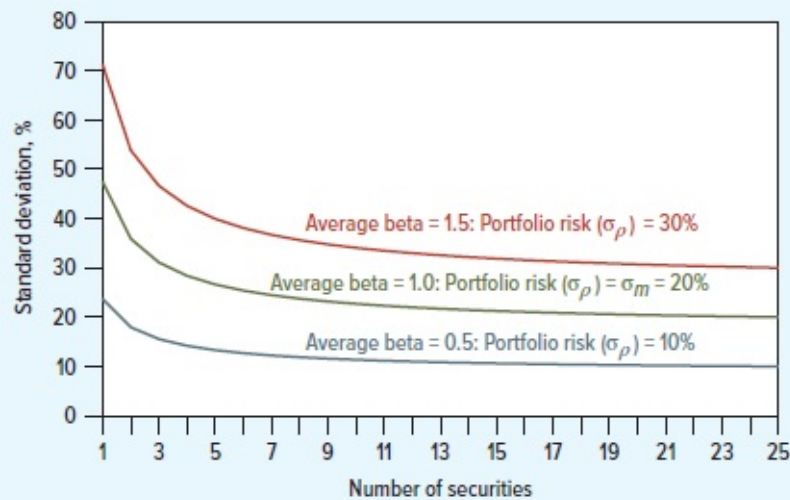
Let's review the two crucial points about security risk and portfolio risk:

The risk of a well-diversified portfolio is given by its market risk.

The market risk of a stock is measured by its beta.

We add a third crucial point: *The beta of a portfolio is the weighted average of the betas of the individual stocks within that portfolio.*² We could give a mathematical proof of this, but we'd much rather give a common-sense one. Remember the intuition for why the total risk (σ_p) of a portfolio is lower than the average of the total risks of the individual stocks. That's because some of the total risk of an individual stock is diversifiable and so goes away when you add it to a portfolio. But since beta measures undiversifiable risk to begin with, there's no diversification effect when adding a stock to a portfolio. So the beta of a portfolio is simply the weighted-average beta of the stocks in the portfolio—no more, no less. The weight for each stock holding is its market value as a percentage of portfolio value.

The beta of a well-diversified portfolio in turn determines its risk. Suppose we construct a portfolio containing a large number of stocks drawn randomly from the whole market. What will we get? Remember that the average stock has a beta of 1.0, so we'd get a portfolio with a beta of 1.0. Because all of the risk of a well-diversified portfolio is market risk, Eq. (8.3) tells us that its standard deviation is $\beta\sigma_M$, that is, $1.0 \times \sigma_M = \sigma_M$. If the standard deviation of the market is 20% (roughly its average for 1900–2020), then the portfolio standard deviation is also 20%. This is shown by the green line in Figure 8.2.



► FIGURE 8.2

The green line shows that a well-diversified portfolio of randomly selected stocks ends up with $\beta = 1$ and a standard deviation equal to the market's—in this case 20%. The upper red line shows that a well-diversified portfolio with $\beta = 1.5$ has a standard deviation of about 30%—1.5 times that of the market. The lower blue line shows that a well-diversified portfolio with $\beta = 0.5$ has a standard deviation of about 10%—half that of the market.

Note: In this figure we assume for simplicity that the total risks of individual stocks are proportional to their market risks.

But suppose we construct the portfolio from stocks with an average beta of 1.5. Then the portfolio beta is 1.5 and its standard deviation is $1.5 \times 20\% = 30\%$.³ A well-diversified portfolio with a beta of 1.5 will end up with 150% of the market's risk. The upper red line in Figure 8.2 shows this case.

Of course, we could repeat the same experiment with stocks with a beta of 0.5 and end up with a well-diversified portfolio half as risky as the market—a standard deviation of 10%. You can see this also in Figure 8.2.

The general point is this: The risk of a well-diversified portfolio is proportional to the portfolio beta, which equals the average beta of the securities included in the portfolio.

8.2 Self-Test

- Applebaum and Banana have betas of 1.0 and 2.0, respectively. Their returns are less than perfectly correlated. You have a portfolio invested equally in each company. Will its beta be 1.5, higher than 1.5, or lower than 1.5?

- b. Given these betas, what correlation coefficient between Applebaum and Banana returns would lead to the lowest portfolio beta?

8-2 The Relationship between Risk and Return

The last section showed that the market risk of a stock is measured by beta. Now we ask one of the most crucial questions in finance: How much extra return should an investor expect for bearing market risk?

Let's go back to the equation of the capital market line from Chapter 7:

$$r_p = r_f + \frac{r_M - r_f}{\sigma_M} \sigma_p \quad (7.4)$$

That equation gave the expected return for an efficient portfolio that combines lending or borrowing and the market portfolio.

Equation (7.4) doesn't apply to individual stocks because they are *not* efficient as standalone assets. Part of their risk can be diversified away, so an investor should not be rewarded for bearing it. Therefore, the capital market line equation doesn't apply to them.

An investor should only be rewarded for bearing a stock's market risk, because only this risk can't be gotten rid of through diversification. Thus, when applying Eq. (7.4) to an individual stock i , we have to replace total risk by market risk only:

$$r_i = r_f + \frac{r_M - r_f}{\sigma_M} (\text{market risk of stock } i)$$

What's the market risk of stock i ? Equation (8.2) shows that it is $\beta_i \sigma_M$. So we have:

$$\begin{aligned} r_i &= r_f + \frac{r_M - r_f}{\sigma_M} (\beta_i \sigma_M) \\ r_i &= r_f + \beta_i (r_M - r_f) \end{aligned}$$

We can also write this in terms of risk premiums by subtracting r_f from each side:

$$r_i - r_f = \beta_i (r_M - r_f) \quad (8.3)$$

Expected risk premium on stock = beta $\times$ expected market risk premium

Equation (8.3) is one of the most famous equations in finance. It's the equation for the **capital asset pricing model**, or **CAPM**, which was derived in the mid-1960s by economists William Sharpe, John Lintner, and Jack Treynor.⁴ It shows a relationship between risk and return that's fundamental to finance. The model's message is both startling and simple. In a

competitive market, the expected risk premium on *any* security or portfolio—not just efficient portfolios—is its beta multiplied by the *market risk premium* $r_M - r_f$. For example, the expected risk premium on an investment with a beta of 0.5 is half the expected risk premium on the market; the expected risk premium on an investment with a beta of 2 is twice the expected risk premium on the market. This is why beta is such a useful measure of risk.

This relationship is plotted in Figure 8.3, which is known as the **security market line (SML)**. Treasury bills have a beta of 0 and a risk premium of 0. The market portfolio has a beta of 1.0 and a risk premium of $r_M - r_f$. The risk premium of any security or portfolio is directly proportional to its beta.

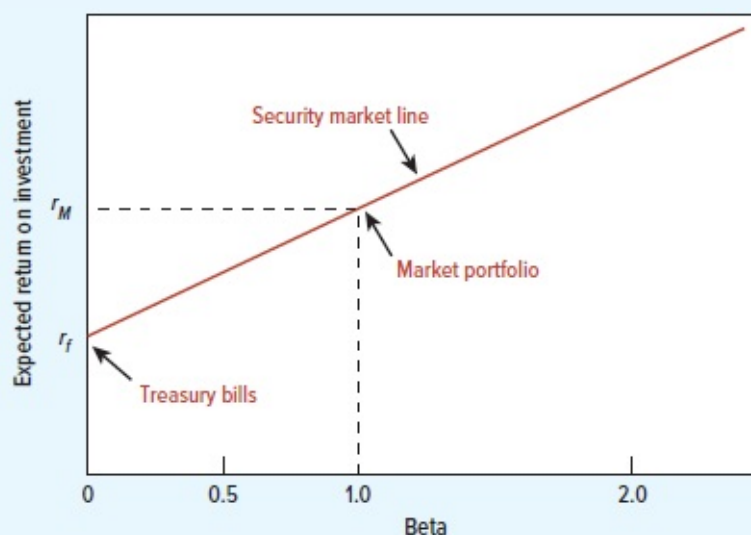


FIGURE 8.3

The capital asset pricing model states that the expected risk premium on each investment is proportional to its beta. This means that each investment should lie on the sloping security market line connecting Treasury bills and the market portfolio.

The CAPM equation is striking for not only what it contains, but also what it does not. It shows that the risk premium of a stock depends on its market risk and only on its market risk. No other term is on the right-hand side of Eq. (8.3). Diversifiable risk should not affect expected returns, nor should any other firm characteristics. We'll revisit this point in Section 8-3, when testing whether the CAPM's predictions hold up in the real world.

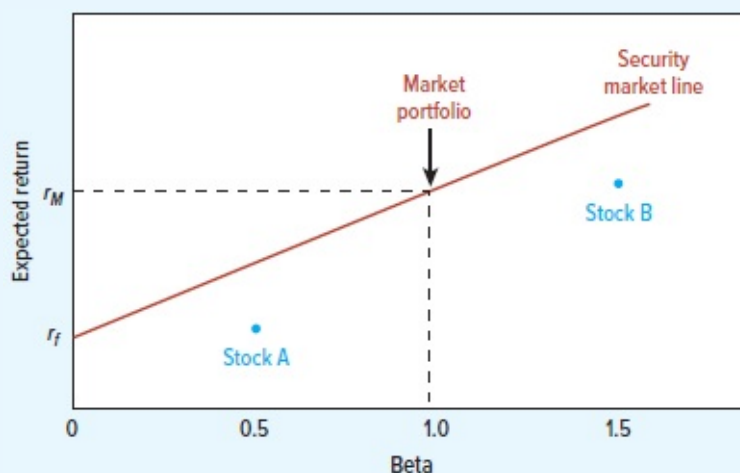
8.3 Self-Test

The risk-free rate is 2%, and the expected return on the market is 8%. What is the expected return of a stock with a beta of (a) 1.5, (b) 0, or (c) $-2/3$?

What If a Stock Did Not Lie on the Security Market Line?

The CAPM is a theory. There is no legislation forcing a stock's expected return to be given by Eq. (8.3). Why might we expect this theory to hold in practice?

Imagine that you encounter stock A in Figure 8.4. Would you buy it? We hope not—unless we were selling it. Buying A alone would leave you undiversified and with an inferior return. Adding it to a diversified portfolio would cure the lack of diversification but not the inferior return. If you want a portfolio with a beta of 0.5, you could get a higher expected return by investing half your money in Treasury bills and half in the market portfolio. If everyone shares your view of the stock's prospects, the price of A will have to fall, increasing the expected return until it matches what you can get on the security market line.



► **FIGURE 8.4**

In equilibrium, no stock can lie below the security market line. For example, instead of buying stock A, investors would prefer to lend part of their money and put the balance in the market portfolio. And instead of buying stock B, they would prefer to borrow and invest in the market portfolio.

What about stock B in Figure 8.4? Would you be tempted by its high return? You shouldn't be. You could get a higher expected return for the same beta by borrowing 50 cents for every dollar of your own money and investing 1.5 times your initial wealth in the market portfolio. B has an inferior return, so no investor would want to buy it on its own or add it to a diversified portfolio. Thus, the price of stock B cannot hold. It will have to fall until the expected return on B rises to equal the expected return on the combination of borrowing and the market portfolio that has the same beta as B.

We have made our point. An investor can always obtain an expected risk premium of $\beta(r_M - r_f)$ by holding a mixture of the market portfolio and the risk-free asset. So in well-functioning markets nobody will hold a stock that offers an expected risk premium of *less* than $\beta(r_M - r_f)$. But what about the other possibility? Are there stocks that offer a higher expected risk premium and therefore lie *above* the security market line in Figure 8.4? Suppose there were. Then those stocks would offer a higher return than other stocks of the same systematic risk. Investors would pile into those stocks, pushing the price up and the expected return down, until they lay on the security market line and offered the appropriate return for their beta. Overall, every stock must lie on the security market line and command an expected risk premium of:

$$r - r_f = \beta (r_M - r_f)$$

The Capital Market Line and the Security Market Line

People often get confused between the security market line and the capital market line. Both lines have expected returns on the vertical axis, but the capital market line has total risk σ_i on the horizontal axis, while the security market line has market risk β_i .

The capital market line applies to *efficient portfolios only*. Inefficient portfolios that bear specific risk, including individual stocks, lie below the line. They carry more risk than efficient portfolios that give the same expected return. But investors are still willing to hold these inefficient stocks because part of their risk is diversifiable. That risk goes away when held as part of a well-diversified portfolio. So, stocks can lie below the capital market line and investors are still willing to hold them.

In contrast, the security market line applies to all stocks and portfolios, efficient or not. That's because the horizontal axis is market risk β_i . Since this risk is undiversifiable, every ounce of it must be rewarded, else an investor would be unwilling to bear it. Thus, the investor needs to be given the risk premium $r_M - r_f$ on the entire amount of market risk β_i . The equation of the security market line, which is the capital asset pricing model, holds for all assets.

In everyday language, being “efficient” is a good thing. We like products or processes that are efficient. In finance also, being efficient is a good thing: We want to get the highest possible return for a given level of risk. However, “efficiency” only matters at a portfolio level. An investor may be perfectly willing to hold inefficient assets, which is why individual assets lie below the capital market line, because these inefficiencies (specific risk) wash away when you hold them as part of a portfolio. An efficient portfolio is comprised of individually inefficient assets.

The Logic behind the Capital Asset Pricing Model

A common job interview question for MBAs is, “Explain the CAPM to your grandparents.” Stepping back from the equations, what’s the logic behind the CAPM? Let’s review the basics.

- Investors like high expected return and low risk. Portfolios that offer the highest expected return for a given level of risk are known as *efficient portfolios*.
- If the investor can lend or borrow at the risk-free rate of interest, one efficient portfolio is better than all the others: the portfolio that offers the highest ratio of risk premium to standard deviation (portfolio T in Figure 7.14). A risk-averse investor will put part of his money in this efficient portfolio and part in the risk-free asset. A risk-tolerant investor may put all her money in this portfolio or she may borrow and put in even more.
- Since everyone holds the same portfolio of stocks, it must be the market portfolio. Everyone holds the market portfolio. Only market risk is relevant.

Now we’ll go back to the risk of individual stocks:

- Investors don’t hold individual stocks, they hold portfolios. So what matters isn’t the total risk of a stock because part of this risk is specific to that stock and diversified away when held as part of a portfolio. What’s left is the systematic risk that’s shared with the rest of the investor’s portfolio.
- Since all investors hold the market portfolio, systematic risk is market risk, the sensitivity of a stock to changes in the market. This sensitivity is known as *beta*.

Intuition: Why Do High Beta and High Returns Go Together?

Why do investors demand a higher return for a stock that has a high beta? Economists would say that high-beta stocks pay off more than low-beta stocks in economic booms, when investors’ marginal utility of more money (i.e., the extra happiness from greater returns) is low, and less in recessions, when marginal utility is high. Investors compensate by requiring a higher expected rate of return from high-beta stocks. The opportunity cost of capital from high-beta investments is likewise higher.

A high-beta stock pays off most when the market is booming. Because investors hold the market, their portfolio does well in a boom, when they’re probably doing well personally and their marginal utility of more money is relatively low. If they’re already able to pay the rent, afford their kids’ school fees, and go on a couple of holidays, then the extra return from the high-beta stock isn’t as valuable to them as it would be in tougher times. In contrast, when the market crashes, investors’ portfolios plummet at the same time when they’re probably not doing so well personally and the marginal utility of more money is relatively high. But if a stock is high beta, then it’s also likely to suffer more in a market crash. The stock may lose

more money precisely at the time when you need it. A high-beta stock is like a fair-weather friend who goes to your garden party when the weather is sunny and everyone else is showing up, but ditches you when it's rainy and you'd really appreciate the company.

In contrast, a low-beta stock is less sensitive to the market. It doesn't do so well when the market is booming and the marginal utility of more money is relatively low. The strength of the low-beta stock is in bad times. That's when your portfolio is underperforming and you really need the money and marginal utility is relatively high. That's when the low-beta stock is likely to deliver more than high-beta stocks. Since it provides valuable insurance against tough times, you're willing to hold it even if its overall return, averaged across booms and recessions, is lower.

When choosing stocks, you don't just care about *how much you get* (the expected return), but *when you get it* (whether returns arise in good or bad times). That's what beta measures.

8.4 Self-Test

Why does a stock's expected return depend only on its market risk β , not its total risk σ ?

Page 232

FINANCE IN PRACTICE

Can Expected Returns Be Negative?

› Let's go back to the CAPM formula:

$$r - r_f = \beta (r_M - r_f)$$

If beta is negative—which Section 8.1 argued is possible—then the expected return on a stock can be *less* than the risk-free rate. Indeed, if beta is sufficiently negative, then the expected return might even be less than zero. Does that make sense? Why would anyone be willing to hold a stock that's expected to lose money?

Because of insurance. Ordinary insurance is a useful analogy here. Let's say you have a house worth \$1 million, and it has a 1% chance of burning down. You're offered an insurance policy that pays out the value of your house if it burns down. The expected payoff to the policy is 1% of \$1 million = \$10,000. If the insurance policy costs \$20,000, many households will still be willing to buy it, even though its expected return is *minus* 50%. (You pay \$20,000 and get back something with an expected value of \$10,000.)

Households haven't miscalculated, nor are they being foolish. They buy the insurance contract because it pays off when their house has burned down and they really need the money.

Applying the Capital Asset Pricing Model

Now that we've understood the logic behind the CAPM, let's now *use* it by plugging in actual numbers. The CAPM requires three inputs: β , r_f and $r_M - r_f$. Let's tackle these in turn:

β : Table 8.1 shows betas for 10 stocks.

r_f : We assume that the interest rate on Treasury bills is 2%.⁵

$r_M - r_f$: We assume that the expected market risk premium is 7%.

Table 8.4 puts these numbers together to give an estimate of the expected return on each stock.⁶ The stock with the highest beta in our sample is U.S. Steel. Our estimate of the expected return from U.S. Steel is 22.9%. The stock with the lowest beta is Newmont. Our estimate of its expected return is just 3.1%.

Stock	Beta (β)	Expected Return $r_f = \beta(r_M - r_f)$
United States Steel	2.98	22.9
Southwest Airlines	1.58	13.0
Amazon	1.55	12.8
Wells Fargo	1.14	10.0
ExxonMobil	1.14	10.0
Johnson & Johnson	0.75	7.3
Tesla	0.50	5.5
Coca-Cola	0.46	5.2
Consolidated Edison	0.31	4.1
Newmont	0.16	3.1

TABLE 8.4 These estimates of the returns expected by investors in February 2020 are based on the capital asset pricing model. We assume 2% for the interest rate r_f and 7% for the expected risk premium $r_M - r_f$.

Pick a stock, and assume a capital investment project with the same risk—same beta—as the stock. The opportunity cost of capital for such a project is calculated in Table 8.4. Given the table's assumptions, the cash flows from a project with the same risk as Southwest Airlines should be discounted at 13%.

We cover the use of the CAPM in capital budgeting in much more detail in Chapter 9.

8-3 Does the CAPM Hold in the Real World?

The CAPM is only a theory. Like any theory, it makes several assumptions. Some of them we've stated explicitly; others we've made implicitly along the way. Let's make them all explicit and in one place:

- Investors choose their portfolios based on expected return and risk, measured by variance of return.
- All investors have the same estimates of the mean returns, variances, and covariances of all assets.
- Investors trade in perfect capital markets. There are no frictions such as taxes, transaction costs, or restrictions on short sales. Investors can both borrow and lend at the same risk-free rate.
- Investors are price takers and cannot influence prices.
- The supply of all assets is fixed.

These assumptions don't hold perfectly in the real world, so the CAPM may seem unrealistic. But the straight lines of subway transit systems, such as the New York subway or the London Underground, are also unrealistic because the subway trains don't go in a straight line between stations. Yet the maps remain practically useful even though they are not fully realistic.

So the key question is this: Is the CAPM a sufficiently valid description of the real world that practitioners can use it in practice? To answer this question, let's look at the data. First, it is indeed the case that asset classes with more market risk command a higher expected return. We saw this in Section 7-1: Stocks offer a higher return than government bonds, which in turn offer a higher return than Treasury bills. This relationship is true in all developed economies over long time periods. Moreover, within the broad category of stocks, shares with high betas earn greater returns than those with low betas.

Second, recall from Section 8-2 that the CAPM is important not only for what it does include, but also for what it does not. The CAPM equation $r_i - r_f = \beta_i(r_M - r_f)$ not only means

that a security's expected return should depend on its beta, but also that it shouldn't depend on anything else. In particular, it shouldn't depend on diversifiable risk. This prediction is confirmed in practice. Stock prices don't increase when two firms merge just to spread their risks; investment companies, which invest in shares of other firms, aren't more highly valued than the underlying shares that they hold. Investors don't pay a premium for actions that reduce diversifiable risk because they can diversify cheaply and easily on their own.

So we start with two victories for the CAPM: Expected returns are positively related to market risk and unrelated to idiosyncratic risk. The CAPM captures these facts in a simple way. That's why nearly three-quarters of financial managers use it to estimate the cost of capital.⁷

But the CAPM actually makes stronger predictions. Not only does it argue that expected returns increase with beta, but it predicts *how much* they increase. An increase in beta of 1.0 should raise the expected return by $r_M - r_f$. Does this stronger prediction hold in the real world?

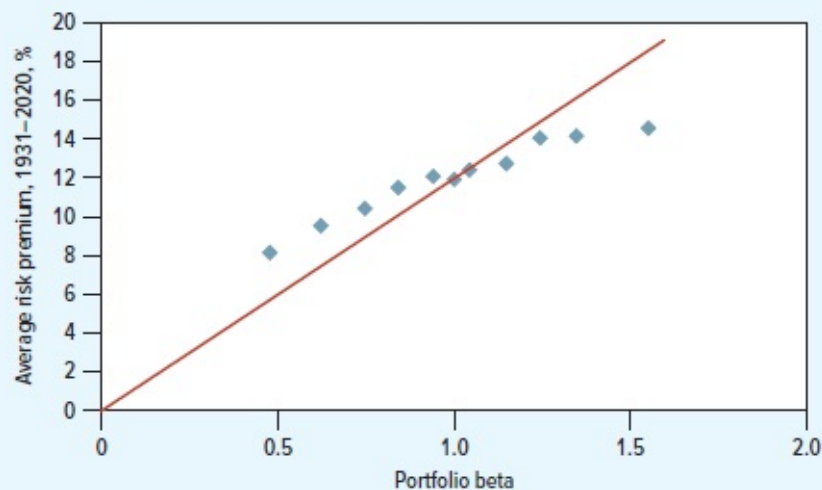
How Large Is the Return for Risk?

Imagine that, in 1931, 10 investors gathered together in a Wall Street bar and agreed to establish investment trust funds for their children. Each investor decided to follow a different strategy. Investor 1 bought the 10% of stocks with the lowest estimated betas; investor 2 chose the 10% with the next-lowest betas; and so on, up to investor 10, who proposed to buy the stocks with the highest betas. At the end of each year, they would re-estimate the betas of all NYSE stocks and reconstitute their portfolios. And so they parted with much cordiality and good wishes.

In time, the 10 investors all passed away, but their children met in early 2020 in the same bar to compare the performance of their portfolios. Figure 8.5 shows how they had fared. Investor 1's portfolio turned out to be much less risky than the market; its beta was only 0.48. However, he also realized the lowest return, 8.1% above the risk-free rate of interest. At the other extreme, the beta of investor 10's portfolio was 1.55, about three times investor 1's beta. But she was rewarded with the highest return, averaging 14.6% a year above the interest rate. So over this 88-year period, returns did indeed increase with beta.

As you can see from Figure 8.5, the market portfolio over the same 89-year period provided an average return of 11.9% above the interest rate⁸ and (of course) had a beta of 1.0. The CAPM predicts that the risk premium should increase in proportion to beta, so that the returns of each portfolio should lie on the upward-sloping security market line in Figure 8.5. Since the market provided a risk premium of 11.9%, investor 1's portfolio, with a beta of 0.48, should have provided a risk premium of 5.7% and investor 10's portfolio, with a beta of 1.55, should have given a premium of 18.5%. You can see that, while high-beta stocks performed better than low-beta stocks, the difference was not as great as the CAPM predicts.

The additional return for a unit of risk was *less* than the 11.9% predicted by the CAPM.

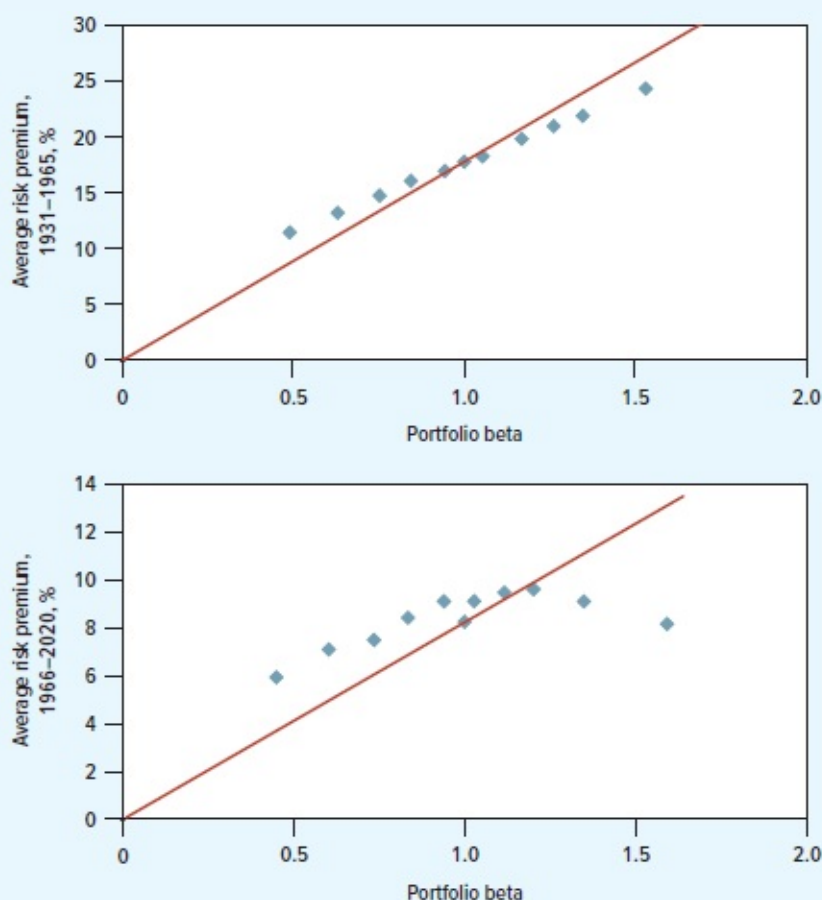


► FIGURE 8.5

The capital asset pricing model states that the expected risk premium from any investment should lie on the security market line. The dots show the actual average risk premiums from portfolios with different betas. The high-beta portfolios generated higher average returns, just as predicted by the CAPM. But the high-beta portfolios plotted below the market line, and the low-beta portfolios plotted above. A line fitted to the 10 portfolio returns would be “flatter” than the security market line.

Source: F. Black, “Beta and Return,” *Journal of Portfolio Management* 20 (Fall 1993), pp. 8–18. Updates courtesy of Adam Kolasinski.

Although Figure 8.5 provides broad support for the CAPM, critics point out that the slope of the line has been particularly flat in the last 50 years. For example, Figure 8.6 shows how our 10 investors fared between 1966 and 2019. Now it is less clear who is buying the drinks. Returns are more or less in line with CAPM predictions, with the important exception of the two highest beta portfolios. Investor 10, who rode the roller coaster of the highest beta portfolio, earned a return that was barely above the market. Of course before 1966, the line was correspondingly steeper. This is also shown in Figure 8.6.



► FIGURE 8.6

The relationship between beta and actual average return has been weaker since the mid 1960s. Stocks with the highest betas have provided poor returns.

Source: F. Black, "Beta and Return," *Journal of Portfolio Management* 20 (Fall 1993), pp. 8–18. Updates courtesy of Adam Kolasinski.

8.5 Self-Test

Suppose investors 9 and 10 had an infallible crystal ball in 1931 that predicted the results shown in Figures 8.5 and 8.6. How could they have improved their investment strategies (if allowed by the rules of their game)? Assume that all CAPM assumptions held exactly.

What is going on in the bottom of Figure 8.6? It is hard to say. Defenders of the CAPM emphasize that it predicts *expected* returns, whereas we can observe only *actual* returns. Maybe investors in high-beta stocks got unlucky after 1966, and happened to receive lower returns than they expected. Perhaps the best we can do is to focus on the longest period for which we have reasonable data. This takes us back to Figures 8.5, which suggests that expected returns do increase with beta, though less rapidly than the CAPM predicts.

Another difficulty is that the market definition used in CAPM tests is a proxy for the market portfolio that investors focus on. As we noted in the last section, the market should ideally include all foreign as well as domestic stocks, all bonds as well as stocks, and all other asset classes. It turns out that using an imperfect market proxy can generate results that seem to disprove the CAPM even when the theory is working perfectly.⁹

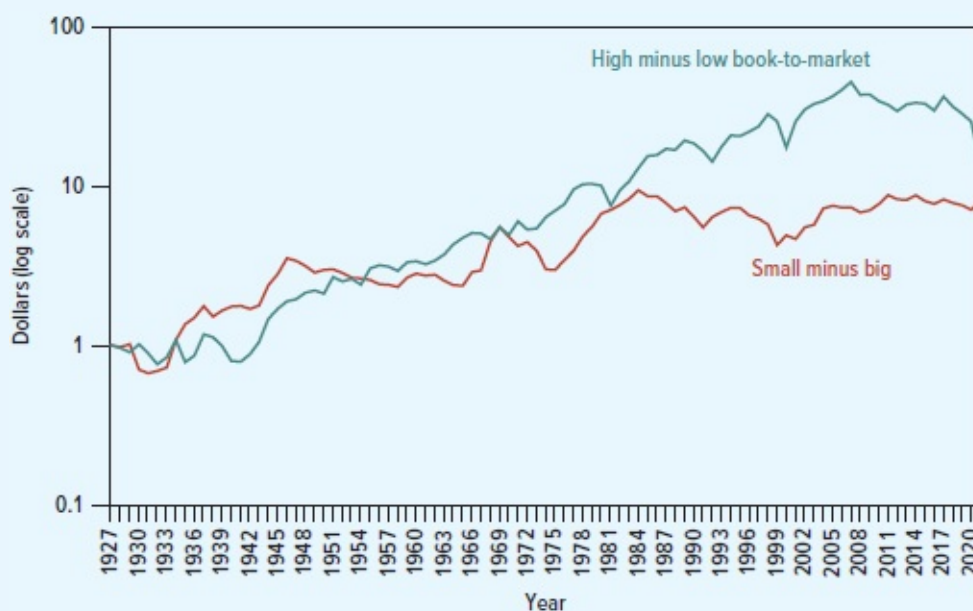
One key assumption of the CAPM is that investors can both borrow and lend at the risk-free rate. Then, a stock's beta doesn't matter because an investor can borrow or lend to get whatever beta he desires. The chicken only invests some of his money in the stock and puts the remainder in the risk-free rate; the lion borrows to invest more than her initial wealth into the stock. But, if some investors can't borrow at the risk-free rate, the only way that they can get a high-beta is to buy high-beta stocks—they can't buy low beta stocks and lever up their position by borrowing. This increases the prices of high-beta stocks, and reduces their return. Thus, high-beta stocks have lower returns than they should do under the CAPM, and low-beta stocks have higher returns than they should do under the CAPM—which is exactly what Figures 8.5 and 8.6 suggest.¹⁰

Indeed, the evidence in Figures 8.5 and 8.6 gives rise to a simple trading strategy: buy low-beta stocks, and sell high-beta stocks. It also means that if you *can* borrow at close to the risk-free rate, the best way to get a high-beta portfolio is not to buy high-beta stocks (they're expensive, because they're sought after by investors who can't borrow) but to buy low-beta stocks and lever up.

This is precisely the strategy pursued by Warren Buffett. He's famous for liking safe companies, and generates a high return by borrowing at very low rates. How? Because he owns several insurance businesses, such as GEICO and National Indemnity. Running an insurance business is effectively borrowing, since you collect the premiums up front and pay out on the policies later. Since his insurance business is very successful, he's able to borrow at exceptional terms.

Are Returns Unrelated to All Other Characteristics?

The CAPM has also come under fire on a second front: Although returns are unrelated to idiosyncratic risk, they're related to other firm characteristics. The red line in Figure 8.7 shows the cumulative difference between the returns on small and large stocks, where size is measured by a stock's market capitalization. You can see that small-cap stocks did not always do well, but over the long haul they have significantly outperformed large-caps. Since the end of 1926, the average annual difference between the returns on the two groups has been 3.1%.



► FIGURE 8.7

The red line shows the cumulative difference between the returns on small-firm and large-firm stocks from 1926 to 2020. The green line shows the cumulative difference between the returns on high book-to-market-value stocks (value stocks) and low book-to-market-value stocks (growth stocks).

Source: Kenneth French's website, mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html.

Now look at the green line in Figure 8.7, which shows the cumulative difference between the returns on value stocks and growth stocks. Value stocks are those with high ratios of book value to market value, and growth stocks have low ratios.¹¹ Since 1926, value stocks have outperformed growth stocks by an average of 4.0% per year.

How seriously is the CAPM damaged by these findings? If you look long and hard at past returns, you're bound to find some strategy that just by chance would have worked in the past. If the returns to the strategy are due to luck—a "spurious correlation"—rather than a risk that investors care about, they should not persist going forward. There's some evidence that this is the case. For example, if you look again at Figure 8.7, you will see that since the mid-1980s, small-firm stocks have underperformed just about as often as they have overperformed.

BEYOND THE PAGE



The CAPM is widely used in corporate finance. It's simple. It makes sense. It seems to generate sensible estimates of costs of capital. On the other hand, the CAPM doesn't hold perfectly in the real world. It's time to consider other theories of risk and return.

8-4 Some Alternative Theories

The capital asset pricing model argues that the only risk that investors are concerned about is market risk and that only one measure of risk (beta) is needed. We now turn to an alternative theory that accommodates several sources of risk.

BEYOND THE PAGE



Arbitrage Pricing Theory

The CAPM begins with an analysis of how investors construct efficient portfolios. Stephen Ross's **arbitrage pricing theory**, or **APT**, comes from a different family entirely. It does not ask which portfolios are efficient. Instead, it starts by *assuming* that each stock's risk premium depends on pervasive macroeconomic “factors,” for example the factors that drive returns on small stocks. Stock returns are assumed to obey the following simple relationship:

$$\text{Return} = a + b_1(r_{\text{factor 1}}) + b_2(r_{\text{factor 2}}) + b_3(r_{\text{factor 3}}) + \dots + \text{noise} \quad (8.4)$$

“Noise” stands for specific, diversifiable risk.

APT theory does not say what the factors are: There could be an oil price factor, an interest-rate factor, and a small stock factor. The return on the market portfolio *might* be one factor, but then again it might not.

Some stocks are more sensitive to a particular factor than other stocks. ExxonMobil would be more sensitive to an oil price factor than, say, Coca-Cola. If factor 1 picks up unexpected changes in oil prices, b_1 will be higher for ExxonMobil.

The APT agrees with the CAPM that there are two sources of risk for an individual stock.

The first is the risks specific to the company. Since specific risk can be eliminated by diversification, it doesn't affect a stock's risk premium. The second is the risk that stems from the macroeconomic factors. These "macro" risks can't be eliminated by diversification and therefore determine expected risk premiums.

The APT states that the expected risk premium on a stock depends on the expected risk premium associated with each factor and the stock's sensitivity to each factor ($b_1, b_2, b_3, \dots$). Thus the formula is

$$\begin{aligned}\text{Expected risk premium} &= r - r_f \\ &= b_1 (r_{\text{factor 1}} - r_f) + b_2 (r_{\text{factor 2}} - r_f) + \dots\end{aligned}\tag{8.5}$$

Notice that this formula makes two statements:

- If all of the b 's in the formula are zero, the expected risk premium is zero. A diversified portfolio that has zero sensitivity to each macroeconomic factor is essentially risk-free (any risk is specific and thus can be diversified away) and therefore must offer the risk-free return. If the portfolio offered a higher return, you could make a risk-free (or "arbitrage") profit by borrowing to buy the portfolio. If it offered a lower return, you could make an arbitrage profit by running the strategy in reverse: You would *sell* the diversified zero-sensitivity portfolio and *invest* the proceeds at the risk-free rate of interest. This explains why the model is known as arbitrage pricing theory.
- The risk premium of a diversified portfolio will vary in direct proportion to its sensitivity to each factor. For example, assume that two portfolios, A and B, are sensitive only to factor 1. If portfolio A has a sensitivity of 0.6 and portfolio B a sensitivity of 0.3, portfolio A has double the sensitivity and so must offer double the risk premium.

Suppose that this wasn't the case, and that A offers a risk premium of 7% but B offers a risk premium of only 3%. Then, an investor who owns B could make an arbitrage profit by selling it and investing the proceeds 50–50 in the risk-free rate and A. The risk of the new portfolio is $50\% \times 0 + 50\% \times 0.6 = 0.3$, the same as before. But the risk premium is $50\% \times 0 + 50\% \times 7\% = 3.5\%$, which is greater than before.

The arbitrage that we have described applies to well-diversified portfolios that have no specific risk. But if the arbitrage pricing relationship holds for all diversified portfolios, it must generally hold for individual stocks, because individual stocks are held in portfolios and their specific risk is diversified away. Each stock must offer an expected return commensurate with its contribution to portfolio risk. In the APT, this contribution depends on the sensitivity of the stock's return to unexpected changes in the macroeconomic factors.

A Comparison of the Capital Asset Pricing Model and Arbitrage Pricing Theory

Both the CAPM and APT claim that a stock's risk premium depends only on its systematic

risk. The CAPM argues that there is only one source of systematic risk—that is, a stock's exposure to the risk of the market portfolio. The APT argues that there could be many sources of systematic risk. The sensitivity to the market portfolio could be one of them, but need not be.

How do the two theories stack up? The APT has some attractive features. For example, the market portfolio that plays such a central role in the CAPM does not feature in arbitrage pricing theory. We do not have to worry about the dangers of using a poor market proxy.

Unfortunately, you win some and lose some. The APT does not tell us what the underlying factors are—unlike the CAPM, which collapses *all* macroeconomic risks into a well-defined *single* factor, the return on the market portfolio.¹² One way to use the APT in practice is to try out plausible factors to see which candidates seem to explain past stock returns. The danger with that approach is that some of these factors will end up significant by chance—their correlation with returns is spurious. If so, investors don't actually require a risk premium for exposure to those factors, and the factors will not predict future returns reliably.

Macroeconomic factors can be measured directly or indirectly. Direct measures might include unexpected changes in energy prices, inflation, or long-term interest rates. Indirect measures use returns on specific portfolios of stocks. Suppose, for example, that the higher historical returns on value stocks are traced back to the stocks' exposure to business-cycle risks. The returns on value stocks could then be used as a proxy for business-cycle risks. That may be their role in the following example.

8.6 Self-Test

What does APT theory say about specific risk? Should a firm's specific risk be included as a factor explaining the expected risk premium?

Page 239

The Three-Factor Model

Look back at the equation for APT. To estimate expected returns, you first need to follow three steps:

Step 1: Identify

a reasonably short list of macroeconomic factors that could affect stock returns.

Step 2: Estimate the expected risk premium on each of these factors ($r_{\text{factor } 1} - r_f$ etc.).

Step 3: Measure the sensitivity of each stock to the factors (b_1, b_2 , etc.).

One way to shortcut this process is to take advantage of the research by Fama and French, which showed that stocks of small firms and those with a high book-to-market ratio have provided above-average returns. This could simply be a coincidence. But there is also

some evidence that these factors are related to company profitability—small and value stocks are less profitable in downturns—and therefore may be picking up risk factors that are left out of the simple CAPM.¹³

If investors do demand an extra return for taking on exposure to these factors, then we can plug these factors into the formula for the APT:

$$r - r_f = b_{\text{market}} (r_{\text{market factor}}) + b_{\text{size}} (r_{\text{size factor}}) + b_{\text{book-to-market}} (r_{\text{book-to-market factor}})$$

This is commonly known as the Fama–French three-factor model. Using it to estimate expected returns is the same as applying the APT. Here is an example.¹⁴

Step 1: Identify the Factors Fama and French have already identified the three factors that appear to determine expected returns. The returns on each of these factors are

Factor	Measured by
Market factor	Return on market index <i>minus</i> risk-free interest rate
Size factor	Return on small-firm stocks <i>less</i> return on large-firm stocks
Book-to-market factor	Return on high book-to-market-ratio stocks <i>less</i> return on low book-to-market-ratio stocks

Step 2: Estimate the Risk Premium for Each Factor We will keep to our figure of 7% for the market risk premium. History may provide a guide to the risk premium for the other two factors. As we saw earlier, between 1926 and 2019, the difference between the annual returns on small and large stocks averaged 3.1% a year. The difference between the returns on stocks with high and low book-to-market ratios averaged 4.0%.

Step 3: Estimate the Factor Sensitivities Some stocks are more sensitive than others to fluctuations in the returns on the three factors. You can see this from the first three columns of numbers in Table 8.5, which show some estimates of the factor sensitivities of 10 industry groups for the 60 months ending in February 2020. For example, an increase of 1% in the return on the book-to-market factor *reduces* the return on pharmaceutical stocks by 0.38% but *increases* the return on oil and gas stocks by 0.49%. In other words, when value stocks (high book-to-market) outperform growth stocks (low book-to-market), computer stocks tend to perform relatively badly and oil and gas stocks do relatively well.

	Three-Factor Model			CAPM	
	Factor Sensitivities			Expected Return ^a	Expected Return ^b
	b_{market}	b_{size}	$b_{\text{book-to-market}}$		
Autos	1.12	0.26	0.31	11.9%	10.2%
Banks	1.18	0.09	0.65	13.1	10.5
Chemicals	1.19	0.26	0.41	12.8	10.7
Computers	1.30	-0.18	0.03	10.5	10.9
Construction	0.96	0.49	0.04	10.4	9.3
Food	0.67	-0.38	0.02	5.6	6.3
Oil and gas	1.17	0.49	0.90	15.3	11.0
Pharmaceuticals	0.91	0.20	-0.38	7.5	8.5
Telecoms	0.78	-0.24	0.12	7.2	7.2
Utilities	0.46	-0.36	-0.08	3.7	4.8

TABLE 8.5 Estimates of expected equity returns for selected industries using the Fama–French three-factor model and the CAPM.

^a The expected return equals the risk-free interest rate plus the factor sensitivities multiplied by the factor risk premiums, that is, $2.0 + (b_{\text{market}} \times 7.0) + (b_{\text{size}} \times 3.1) + (b_{\text{book-to-market}} \times 4.0)$.

^b Estimated as $r_f + \beta(r_m - r_f)$, that is, $r_f + \beta \times 7.0$. Note that we used *simple* regression to estimate β in the CAPM formula. This beta may, therefore, be different from b_{market} that we estimated from a *multiple* regression of stock returns on the three factors.

Source: The industry indexes are value-weighted indexes from Kenneth French's website, mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.xhtml.

Once you have estimated the factor sensitivities, multiply each of them by the expected factor return and add up the results. For example, the expected risk premium on oil and gas stocks is:

$$r - r_f = (1.17 \times 7.0) + (0.49 \times 3.1) - (0.90 \times 4.0) = 13.3\%$$

To calculate the expected return we need to add on the risk-free interest rate, which we assume to be 2%. This yields an expected return of $2.0 + 13.3 = 15.3\%$.

Compare this figure with the expected return estimate using the CAPM (the final column of Table 8.5). The three-factor model provides a higher estimate of the expected return for oil and gas stocks. Why? Largely because they are value stocks with a high exposure (0.90) to the book-to-market factor. The three-factor model produces a higher expected return for value stocks, but it produces a lower figure for growth stocks such as pharmaceutical companies that have low book-to-market ratios.¹⁵

This Fama–French APT model is not widely used in practice to estimate a company's cost of equity. The model requires three betas and three risk premiums instead of one beta

and one market risk premium in the CAPM. Also the three Fama–French betas are not as easy to predict and interpret as the CAPM beta, which measures market risk, and there’s a logical reason for why investors should be compensated for bearing market risk. The logic for using Fama–French risk factors is sometimes elusive.

The Fama–French model finds its widest use as a way of measuring the performance of *investors*, such as mutual funds, pension funds, and other professionally managed portfolios. If a portfolio manager “beats the S&P,” it may be because he simply held small stocks at a time that small stocks soared, not because of any skill in picking individual stocks. Since an investor can buy a passive small-stock index fund, she would not wish to pay a portfolio manager to do so. She can evaluate the manager’s performance by estimating the portfolio’s b_{market} , b_{size} , and $b_{\text{book-to-market}}$ and then checking whether the portfolio return is better than the return on a computer-managed portfolio with the same exposures to the Fama–French factors.



KEY TAKEAWAYS

- **A stock’s market risk is measured by its beta.**

$$\beta_i = \sigma_{iM} / \sigma_M^2 \quad (8.1)$$

where σ_{iM} is the *covariance* between the return on stock i and the market return and σ_M^2 is the variance of the market return. Beta describes how sensitive the stock’s return is to unexpected market returns. The average beta of all stocks is 1.0. The beta of a portfolio is the weighted average of the betas of the individual stocks within that portfolio.

- **The capital asset pricing model (CAPM) states that stocks are priced to offer expected risk premiums that are proportional to beta.**

$$\begin{aligned} \text{Expected risk premium} &= \text{beta} \times \text{market risk premium} \\ r - r_f &= \beta (r_m - r_f) \end{aligned} \quad (8.3)$$

The CAPM says that beta is a complete and sufficient measure of risk. Diversifiable risk does not matter. Other attributes—for example, whether the stock is small or large-cap—do not matter.

- **Real-world evidence supports the CAPM, but with important exceptions.** Market

risk clearly matters in practice. Diversifiable risk does not. These facts are good news for the CAPM. Average past stock returns increase as beta increases but by less than the CAPM predicts. The relationship between beta and average returns is especially weak for the highest-beta stocks. Stocks of small-cap companies, and stocks with high book values relative to market values, appear to have risks not captured by the CAPM.

- **Arbitrage pricing theory (APT).** The APT theory states that the expected risk premium on a stock should depend on the stock's exposure to several macroeconomic factors that affect stock returns:

$$\text{Expected risk premium} = b_1 (r_{\text{factor 1}} - r_f) + b_2 (r_{\text{factor 2}} - r_f) + \dots \quad (8.5)$$

Here b 's represent the stock's sensitivities to the factors, and $r_{\text{factor}} - r_f$ is the risk premium demanded by investors who are exposed to this factor.

Arbitrage pricing theory does not say what the factors are. It asks for economists to hunt for unknown game with their statistical toolkits. Fama and French have suggested three factors:

- The market risk premium.
- The difference between the return on small- and large-cap stocks.
- The difference between the return on stocks with high book-to-market ratios and stocks with low book-to-market ratios.

In the Fama–French three-factor model, the expected risks premium depends on the exposures to the three factors.

Page 242

The CAPM and APT theories both have their fan clubs. However, all financial economists agree on two basic ideas: (1) Investors require extra expected return for taking on risk, and (2) they are concerned predominantly with the risk that they cannot eliminate by diversification.



FURTHER READING

A number of textbooks on portfolio selection explain both Markowitz's original theory and some ingenious simplified versions. See, for example,

E. J. Elton, M. J. Gruber, S. J. Brown, and W. N. Goetzmann: *Modern Portfolio Theory and Investment Analysis*, 9th ed. (New York: John Wiley & Sons, 2014).

The literature on the capital asset pricing model is enormous. There are dozens of published texts of the capital asset pricing model. Fischer Black's paper is a very readable example. Discussions of the theory tend to be more uncompromising. Two excellent but advanced examples are Campbell's survey paper and Cochrane's book.

- F. Black, "Beta and Return," *Journal of Portfolio Management* 20 (Fall 1993), pp. 8–18.
- J. Y. Campbell, "Asset Pricing at the Millennium," *Journal of Finance* 55 (2000), pp. 1515–1567.
- J. H. Cochrane, *Asset Pricing*, revised ed. (Princeton, NJ: Princeton University Press, 2009).



PROBLEM SETS



connect

Select problems are available in McGraw-Hill's *Connect*.

Please see the preface for more information.

1. **Stock betas (S8.1)** What is the beta of each of the stocks shown in Table 8.6?

Stock Return if Market Return Is:		
	– 5	+ 5
A	– 5	+ 5
B	–20	+20
C	–8	+8
D	+5	+5
E	+10	–10

TABLE 8.6 Stock betas. See Problem 1.

2. **Stock betas (S8.1)** There are few, if any, real companies with negative betas. But suppose you found one with $\beta = -0.25$.
- How would you expect this stock's rate of return to change if the overall market rose by an extra 5%? What if the market fell by an extra 5%?
 - You have \$1 million invested in a well-diversified portfolio of stocks. Now you receive an additional \$20,000 bequest. Which of the following actions will yield the safest overall portfolio return?
 - Invest \$20,000 in Treasury bills (which have $\beta = 0$).
 - Invest \$20,000 in stocks with $\beta = 1$.
 - Invest \$20,000 in the stock with $\beta = -0.25$.

Explain your answer.

3. **Portfolio betas (S8.1)** A portfolio contains equal investments in 10 stocks. Five have a beta of 1.2; the remainder have a beta of 1.4. What is the portfolio beta?

Page 243

- a. 1.3.
 - b. Greater than 1.3 because the portfolio is not completely diversified.
 - c. Less than 1.3 because diversification reduces beta.
- 4. Portfolio betas (S8.1)** Suppose the standard deviation of the market return is 20%.
- a. What is the standard deviation of returns on a well-diversified portfolio with a beta of 1.3?
 - b. What is the standard deviation of returns on a well-diversified portfolio with a beta of 0?
 - c. A well-diversified portfolio has a standard deviation of 15%. What is its beta?
 - d. A poorly diversified portfolio has a standard deviation of 20%. What can you say about its beta?
- 5. Portfolio betas (S8.1)** Refer to Table 8.1.
- a. What is the beta of a portfolio that has 40% invested in ExxonMobil and 60% in Newmont?
 - b. Would you invest in this portfolio if you had no superior information about the prospects for these stocks? Devise an alternative portfolio with the same expected return and less risk.
 - c. Now repeat parts (a) and (b) with a portfolio that has 40% invested in Coca-Cola and 60% in Amazon.
- 6. CAPM (S8.2)** True or false? Explain or qualify as necessary.
- a. Investors demand higher expected rates of return on stocks with more variable rates of return.
 - b. The CAPM predicts that a security with a beta of 0 will offer a zero expected return.
 - c. An investor who puts \$10,000 in Treasury bills and \$20,000 in the market portfolio will have a beta of 2.0.
 - d. The CAPM predicts that investors demand higher expected rates of return from stocks with returns that are highly exposed to business-cycle risk.
 - e. Investors demand higher expected rates of return from stocks with returns that are very sensitive to fluctuations in the stock market.
- 7. CAPM (S8.2)** True or false?
- a. The CAPM implies that if you could find an investment with a negative beta, its expected return would be less than the interest rate.
 - b. The expected return on an investment with a beta of 2.0 is twice as high as the expected return on the market.
 - c. If a stock lies below the security market line, it is undervalued.
- 8. CAPM (S8.2)** Suppose that the Treasury bill rate is 6% rather than the 2% assumed

in earlier examples. Assume that the expected return on the market stays at 9%. Use the betas in Table 8.1.

- a. Calculate the expected return from Johnson & Johnson.
 - b. Find the highest expected return that is offered by one of these stocks.
 - c. Find the lowest expected return that is offered by one of these stocks.
 - d. Would U.S. Steel offer a higher or lower expected return if the interest rate were 6% rather than 2%? Assume that the expected market return stays at 9%.
 - e. Would Coca-Cola offer a higher or lower expected return if the interest rate were 6%?
9. **CAPM (S8.2)** The Treasury bill rate is 4%, and the expected return on the market portfolio is 12%. Using the CAPM:
- a. Draw a graph similar to Figure 8.3 showing how the expected return varies with beta.
 - b. What is the risk premium on the market?
 - c. What is the required return on an investment with a beta of 1.5?
 - d. If an investment with a beta of 0.8 offers an expected return of 9.8%, does it have a positive NPV?
 - e. If the market expects a return of 11.2% from stock X, what is its beta?
10. **Security market line (S8.2)** Briefly explain the difference between the security market line and the capital market line. *Hint:* One applies to efficient portfolios only.
11. **Evidence for or against the CAPM (S8.3)** True or false?
- a. Long-run average rates of return have been higher for high-beta stocks than for low-beta stocks.
 - b. Low-beta stocks have earned lower rates of return than the CAPM predicts.
 - c. The CAPM predicts that investors do not bid up the values of diversified firms. This prediction is confirmed in practice.
 - d. Small-cap stocks have earned higher rates of return than the CAPM predicts.
 - e. The historical performance of small-cap stocks does not disprove the CAPM, which allows for factors in addition to the market factor.
12. **APT (S8.4)** Consider a three-factor APT model. The factors and associated risk premiums are

Factor	Risk Premium (%)
Change in gross national product (GNP)	+5
Change in energy prices	-1
Change in long-term interest rates	+2

Calculate expected rates of return on the following stocks. The risk-free interest rate is 7%.

- a. A stock whose return is uncorrelated with all three factors.
- b. A stock with average exposure to each factor ($b = 1$ for each).
- c. A pure-play energy stock with high exposure to the energy factor ($b = 2$) but zero exposure to the other two factors.
- d. An aluminum company stock with average sensitivity to changes in interest rates and GNP, but negative exposure of $b = -1.5$ to the energy factor. (The aluminum company is energy-intensive and suffers when energy prices rise.)

13. **APT (S8.4)** Some true or false questions about the APT:

- a. The APT factors cannot reflect diversifiable risks.
- b. The market rate of return cannot be an APT factor.
- c. There is no theory that specifically identifies the APT factors.
- d. The APT model could be true but not very useful, for example, if the relevant factors change unpredictably.

14. **APT (S8.4)** Consider the following simplified APT model:

Factor	Expected Risk Premium (%)
Market	6.4
Interest rate	-0.6
Yield spread	5.1

Calculate the expected return for the following stocks. Assume $r_f = 5\%$.

	Factor Risk Exposures		
	Market	Interest Rate	Yield Spread
Stock	(b_1)	(b_2)	(b_3)
P	1.0	-2.0	-0.2
P ²	1.2	0	0.3
P ³	0.3	0.5	1.0

15. **APT (S8.4)** Look again at Problem 14. Consider a portfolio with equal investments in stocks P, P², and P³.

- a. What are the factor risk exposures for the portfolio?
- b. What is the portfolio's expected return?

16. **Three-factor model (S8.4)** The following table shows the sensitivity of four stocks

to the three Fama–French factors. Estimate the expected return on each stock assuming that the interest rate is 2%, the expected risk premium on the market is 7%, the expected risk premium on the size factor is 3.0%, and the expected risk premium on the book-to-market factor is 4.5%.

	Ford	Walmart	Citigroup	Apple
Market	1.24	0.41	1.52	1.25
Size	−0.07	−0.47	−0.01	−0.67
Book-to-market	0.28	−0.25	0.85	−0.72

CHALLENGE

- 17. Evaluating investment performance (S8.4)** Between 2008 and 2017, the returns on Microfund averaged 10% a year. In his 2017 discussion of performance, the fund president noted that this was 2.5% a year better than the return on the U.S. market, a result that he attributed to the fund’s strategy of buying only stocks with outstanding management.

The following table shows the returns on the market, the size and book-to-market factors, and the interest rate during this period:

	Market Risk Premium	Return on Size Factor	Return on Book-to-Market Factor	Interest Rate
2008	−38.34	3.26	0.97	1.60
2009	28.26	9.28	−9.14	0.10
2010	17.37	13.77	−5.17	0.12
2011	0.44	−6.04	−8.41	0.04
2012	16.28	−1.22	9.89	0.06
2013	35.2	7.35	1.54	0.02
2014	11.7	−7.75	−1.65	0.02
2015	0.07	−3.73	−9.48	0.02
2016	13.3	6.66	23.33	0.20
2017	21.5	−4.85	−13.85	0.80

The fund had marketed itself as a way to invest in small and medium-sized stocks, and this was reflected in a beta relative to the size factor of 1.4. It had also traditionally adopted a conservative approach to risk with an estimated market beta of 0.9. The fund’s beta relative to the book-to-market factor was −0.4. Evaluate the performance of the fund during this period.

- 18. APT (S8.4)** The following question illustrates the APT. Imagine that there are only two pervasive macroeconomic factors. Investments X, Y, and Z have the following sensitivities to these two factors:

Investment	b_1	b_2
X	1.75	0.25
Y	-1.00	2.00
Z	2.00	1.00

We assume that the expected risk premium is 4% on factor 1 and 8% on factor 2. Treasury bills obviously offer zero risk premium.

- According to the APT, what is the risk premium on each of the three stocks?
- Suppose you buy \$200 of X and \$50 of Y and sell \$150 of Z. What is the sensitivity of your portfolio to each of the two factors? What is the expected risk premium?
- Suppose you buy \$80 of X and \$60 of Y and sell \$40 of Z. What is the sensitivity of your portfolio to each of the two factors? What is the expected risk premium?
- Finally, suppose you buy \$160 of X and \$20 of Y and sell \$80 of Z. What is your portfolio's sensitivity now to each of the two factors? And what is the expected risk premium?
- Suggest two possible ways that you could construct a fund that has a sensitivity of 0.5 to factor 1 only. (*Hint:* One portfolio contains an investment in Treasury bills.) Now compare the risk premiums on each of these two investments.
- Suppose that the APT did *not* hold and that X offered a risk premium of 8%, Y offered a premium of 14%, and Z offered a premium of 16%. Devise an investment that has zero sensitivity to each factor and that has a positive risk premium.



SOLUTIONS TO SELF-TEST QUESTIONS

8.1 $\beta_i = \sigma_{im} / \sigma_M^2 = 0.03 / 0.2^2 = 0.75$

8.2

- 1.5. The beta of a portfolio is the (weighted) average of the betas of the individual securities.
- The portfolio beta is a weighted average of the individual betas regardless of correlation.

8.3

a. $2\% + 1.5 \times (8\% - 2\%) = 11\%$

b. 2%

c. $2\% - \frac{2}{3} \times (8\% - 2\%) = -2\%$

- 8.4** Investors should not expect compensation for a stock's total risk σ because part of that risk can be diversified away. They should only expect compensation for undiversifiable risk, which is captured by β .
- 8.5** Investors 9 and 10 could hold the market portfolio, then borrow to bring their portfolio betas up to the betas of stock portfolios 9 and 10. Their average return would move up to the security market line. Implementing this strategy in real life would have been extremely difficult, however, as we point out later in this section.
- 8.6** No. APT theory assumes that all specific risk—all risk unrelated to the factors—is eliminated by diversification.



FINANCE ON THE WEB

You can download data for the following questions from finance.yahoo.com or any financial website that provides stock data.

1. Find a low-risk stock—Walmart or Kellogg would be a good candidate. Use monthly returns for the most recent three years to confirm that the beta is less than 1.0. Now estimate the annual standard deviation for the stock and the S&P index, and the correlation between the returns on the stock and the index. Forecast the expected return for the stock, assuming the CAPM holds, with a market return of 12% and a risk-free rate of 5%.
 - a. Plot a graph like Figure 8.5 showing the combinations of risk and return from a portfolio invested in your low-risk stock and the market. Vary the fraction invested in the stock from 0 to 100%.
 - b. Suppose that you can borrow or lend at 5%. Would you invest in some combination of your low-risk stock and the market, or would you simply invest in the market? Explain.
 - c. Suppose that you forecasted a return on the stock that is 5 percentage points higher than the CAPM return used in part (b). Redo parts (a) and (b) with the higher forecasted return.
 - d. Find a high-risk stock and redo parts (a) and (b).
2. Recalculate the betas for the stocks in Table 8.1 using the latest 60 monthly returns.

Recalculate expected rates of return from the CAPM formula, using a current risk-free rate and a market risk premium of 7%. How have the expected returns changed from Table 8.1?

¹We show estimated betas for five years before the start of the pandemic. As the accompanying app below shows, betas for some stocks changed dramatically during the pandemic.

²As a result, adding a low-beta stock to your portfolio lowers the risk of the overall portfolio. However, this is not “diversification,” which refers specifically to the removal of specific risk. Instead, it is “de-risking,” which refers to the removal of market risk. Investing more of your wealth into Treasury bills is another way to de-risk.

³A 500-stock portfolio with $\beta = 1.5$ still has some specific risk, because it is concentrated by industry and in growth rather than income stocks. Its actual standard deviation would be a bit higher than 30%. If that worries you, relax; you can construct a fully diversified portfolio with a beta of 1.5 by borrowing and investing in the market portfolio.

⁴W. F. Sharpe, “Capital Asset Prices: A Theory of Market Equilibrium under Conditions of Risk,” *Journal of Finance* 19 (1964), pp. 425–442; and J. Lintner, “The Valuation of Risk Assets and the Selection of Risky Investments in Stock Portfolios and Capital Budgets,” *Review of Economics and Statistics* 47 (1965), pp. 13–37. Treynor’s article has not been published.

⁵ Notice that it is the current interest rate that is relevant. The level of interest rates in the past has nothing to do with the returns that investors expect today.

⁶ These expected returns are not the same as the forecasts assumed in Table 7.5 to generate efficient portfolios. Those forecasts were the opinions of a fictitious investor who has not discovered the CAPM.

⁷ See J. R. Graham and C. R. Harvey, “The Theory and Practice of Corporate Finance: Evidence from the Field,” *Journal of Financial Economics* 60 (2001), pp. 187–243. A number of the managers surveyed reported using more than one method to estimate the cost of capital. Seventy-three percent used the capital asset pricing model, while 39% stated they used the average historical stock return and 34% used the capital asset pricing model with some extra risk factors.

⁸ In Figure 8.5, the stocks in the “market portfolio” are weighted equally. Since the stocks of small firms have provided higher average returns than those of large firms, the risk premium on an equally weighted index is higher than on a value-weighted index. This is one reason for the difference between the 11.9% market risk premium in Figure 8.5 and the 7.7% premium reported in Table 7.1. Also, our 10 investors were lucky to set up their trust funds in 1931, just before stock prices rebounded from the Great Crash of 1929.

⁹ This point is known as the Roll critique: R. Roll, “A Critique of the Asset Pricing Theory’s Tests Part 1: On Past and Potential Testability of the Theory,” *Journal of Financial Economics* 4 (1977), pp. 129–176.

¹⁰ The supply of high-beta stocks in the market portfolio is fixed. If risk-tolerant investors are determined to hold more than the market proportion of high-beta stocks, then conservative investors have to hold less than the market proportion. The risk-tolerant investors have to bid up the prices of high-beta stocks in order to convince the conservative investors to sell them.

¹¹ Fama and French calculated the returns on portfolios designed to measure the size and value effects. See E. F. Fama and K. R. French, “The Cross-Section of Expected Stock Returns,” *Journal of Finance* 47 (1992), pp. 427–465. When calculating the returns on these portfolios, Fama and French control for differences in firm size when comparing stocks with low and high book-to-market ratios. Similarly, they control for differences in the book-to-market ratio when comparing small- and large-firm stocks. For details of the methodology and updated returns on the size and book-to-market factors see Kenneth French’s website (mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html).

¹² The CAPM does not deny the importance of “macro” risks. Macro risks move the market. The CAPM says that fluctuations in the market portfolio incorporate the effects of all relevant macro risks. The investor can worry about a single market risk measure instead of the macro risks separately.

¹³ E. F. Fama and K. R. French, “Size and Book-to-Market Factors in Earnings and Returns,” *Journal of Finance* 50 (1995), pp. 131–155.

¹⁴ The three-factor model was first used to estimate the cost of capital for different industry groups by Fama and French. See E. F. Fama and K. R. French, “Industry Costs of Equity,” *Journal of Financial Economics* 43 (1997), pp. 153–193. Fama and French emphasize the imprecision in using either the CAPM or an APT-style model to estimate the returns that investors expect.

¹⁵ A stock’s factor sensitivities depend on the *covariances* of the stock’s return with the factor returns, not on the stock’s market capitalization or other attributes. For example, a large-cap stock can *behave* like a portfolio of small stocks; its return can covary positively with the return on the small-stock portfolio. Returns on some growth stocks can covary more positively with value stocks than with growth stocks generally. For further details, see K. Daniel, and S. Titman, “Evidence on the Characteristics of Cross-Sectional Returns,” *Journal of Finance* 52 (1997), pp. 1–33.